

Lecture 05

Matrix Operation I: Multiplication and Transpose

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群报数会员版

Today's Lecture: Outline

Today ... Matrix Multiplication!

Outline:

1. Matrix multiplication: Properties
2. Matrix multiplication and transpose

After this lecture, you should be able to

1. Apply the **properties of matrix multiplication**
2. Apply **transpose in matrix operations**
3. Tell various forms of vector-vector products

Recall: Matrix-Vector Product

Definition 3.1 (Matrix-vector product definition)

Let $u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and A be an $m \times n$ matrix $A = (a_{ij})$, where $i = 1, \dots, m$; $j = 1, \dots, n$,

The matrix-vector product Au is an $m \times 1$ column vector:

$$Au = \begin{bmatrix} \sum_{j=1}^n a_{1j}u_j \\ \sum_{j=1}^n a_{2j}u_j \\ \vdots \\ \sum_{j=1}^n a_{mj}u_j \end{bmatrix}.$$

Remark: Dimension checking

$$\begin{matrix} A & u & = & v \\ m \times n & n \times 1 & & m \times 1 \end{matrix}$$

Example:

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 3 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times (-1) + 1 \times 6 \\ 3 \times 5 + 4 \times (-1) + 3 \times 6 \end{bmatrix}$$

$2 \times 3 \qquad 3 \times 1$

Part 1 Matrix Multiplication: Definition

15 mins

Matrix Times Matrix: Possibility?

Can a matrix be multiplied by a matrix?

Extra requirement: want the outcome to be a matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}$$

What can “A times B” be?

There are multiple types of products;
we introduce the most powerful and popular one in class.

Matrix Times Matrix v.s. Matrix Times Vector

Can a matrix be multiplied by a matrix?

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 \\ 8 \end{bmatrix} \quad A \cdot C = \begin{bmatrix} 1 \times 2 + 2 \times 8 \\ 3 \times 2 + 4 \times 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} \quad A \cdot B = \begin{bmatrix} 1 \times 2 + 2 \times 8 & 1 \times 1 + 2 \times 7 \\ 3 \times 2 + 4 \times 8 & 3 \times 1 + 4 \times 7 \end{bmatrix}$$

What can "A times B" be?

To be consistent with matrix-vector product, choose
"row-times-column" definition.

Matrix Times Matrix

$$\begin{array}{ccc}
 AB = C & & \\
 \downarrow \quad \downarrow & & \downarrow \\
 m \times n & n \times k & m \times k
 \end{array}$$

Definition 5.1 (Matrix multiplication)

Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$. The product $C = AB$ is an $m \times k$ matrix

where each entry is given by:

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

$b_{kj}, k=1, \dots, n$. j -th column of B

$a_{ik}, k=1, 2, \dots, n$: i -th row of A

where a_{ik} is the (i, k) -th entry of A and b_{kj} is the (k, j) -th entry of B .

Examples:

Rows * columns

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times (-2) & 1 \times (-2) + 2 \times 1 \\ 3 \times 5 + 4 \times (-2) & 3 \times (-2) + 4 \times 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 3 & 4 & 0 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 1 & 5 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 \times 5 + 2 \times 1 + 4 \times (-2) & 1 \times (-2) + 2 \times 5 + 4 \times 1 \\ 3 \times 5 + 4 \times 1 + 0 \times (-2) & 3 \times (-2) + 4 \times 5 + 0 \times 1 \end{bmatrix}$$

2×3 3×2 2×2

Remark: AB exists iff # of columns of A equal to # of rows of B .

The dimensions have to match!

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Matrix Multiplied by a Matrix

Exercise (after class)

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 & 6 \\ 0 & -4 & 1 & 2 & 3 \\ -5 & 1 & 2 & -3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 & 2 & 1 \\ -1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 3 \\ 6 & 4 & -1 & 2 \\ 1 & -2 & 3 & 0 \end{bmatrix}$$

$$AB =$$

$$\begin{bmatrix} 28 & 17 & 20 & 10 \\ 20 & -13 & -3 & -1 \\ -18 & -44 & 12 & -3 \end{bmatrix}$$

Property: Multiplying Each Column

Property Let $A \in \mathbb{R}^{m \times n}$, $B = [\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_r] \in \mathbb{R}^{n \times r}$. The product AB can be computed by:

$$AB = [A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_r].$$

Example

$$A = \begin{bmatrix} 1 & 2 & -1 & 4 & 6 \\ 0 & -4 & 1 & 2 & 3 \\ -5 & 1 & 2 & -3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 & 2 & 1 \\ -1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 3 \\ 6 & 4 & -1 & 2 \\ 1 & -2 & 3 & 0 \end{bmatrix}$$

Then

$$AB = \left[A \begin{bmatrix} 1 \\ -1 \\ 1 \\ 6 \\ 1 \end{bmatrix}, A \begin{bmatrix} 6 \\ 4 \\ 1 \\ 4 \\ -2 \end{bmatrix}, A \begin{bmatrix} 2 \\ 3 \\ 2 \\ -1 \\ 3 \end{bmatrix}, A \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 0 \end{bmatrix} \right] = \begin{bmatrix} 28 & 17 & 20 & 10 \\ 20 & -13 & -3 & -1 \\ -18 & -44 & 12 & -3 \end{bmatrix}$$

Principle: Dimension Match

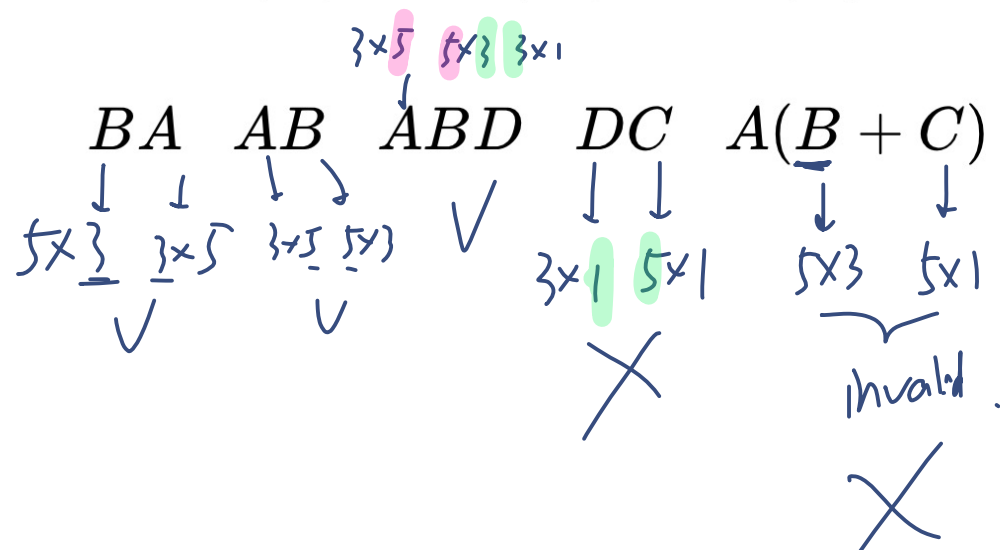
Most important principle of matrix multiplication:
Dimensions must match!

Valid multiplication: dimensions match.

Invalid multiplication: dimension do NOT match.

Exercise

A is 3 by 5, B is 5 by 3, C is 5 by 1, and D is 3 by 1. Which of these matrix operations are allowed?



Identity Matrix

Definition: An identity matrix $I_n \in \mathbb{R}^{n \times n}$ is the following:

$$I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \cdots & \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Sometimes, simply written as I .

Example: $I_3 = \begin{bmatrix} \mathbf{1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix}$

Matrix Multiplication: Special Matrices

Property Let $A \in \mathbb{R}^{m \times n}$, $I \in \mathbb{R}^{n \times n}$ Is an identity matrix
 $AI = A$?

Example:

$$A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AI = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix} = A$$

In-class exercise: 1 min.

Property Multiplied by a zero matrix?

$$A \mathbf{0} = \mathbf{0}$$

Part 2 Matrix Multiplication: Properties

20 mins

Matrix Multiplication Properties (with bugs)

Properties Let $A \in \mathbb{R}^{m \times n}$, $B, C \in \mathbb{R}^{n \times l}$, $\alpha \in \mathbb{R}$

(1) $A(B + C) = AB + AC$ Left Distributive

(2) $(B + C)A = BA + CA$ Right Distributive

(3) $\alpha(AB) = (\alpha A)B = A(\alpha B)$ Scalar Associative

(4) $(AB)C = A(BC)$ Associative

$\downarrow$ $\downarrow$
 $n \times l$ $n \times l$

Any Bugs?

Matrix Multiplication Properties

Properties Suppose A, B, C are matrices with proper dimensions, $\alpha \in \mathbb{R}$

$$(1) \quad A(B + C) = AB + AC$$

Left Distributive

$$(2) \quad (B + C)A = BA + CA$$

Right Distributive

$$(3) \quad \alpha(AB) = (\alpha A)B = A(\alpha B)$$

Scalar Associative

$$(4) \quad (AB)C = A(BC)$$

Associative

In (2): $B, C \in \mathbb{R}^{m \times n}, A \in \mathbb{R}^{n \times k}$.

In (1), (3): $A \in \mathbb{R}^{m \times n}, B, C \in \mathbb{R}^{n \times k}$.

In (4), $A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times k}, C \in \mathbb{R}^{k \times p}$.

Missing Property of Matrix Multiplication

You might be tired of seeing these properties over and over again.

Of course, these properties **should hold for any product**, right?

Question:

What property does matrix multiplication NOT have?

[Compared to: scalar product; vector inner product]

Matrix Multiplication: AB and BA are NOT Equal

1. AB exists does not imply that BA exists.

Example:

$$\begin{array}{cc} A & B \\ 2 \times 3 & 3 \times 5 \\ & \checkmark \end{array}$$

$$\begin{array}{cc} B & A \\ 3 \times 5 & 2 \times 3 \\ = & = \\ \times & \end{array}$$

2. Even if both AB and BA exist, they are generally not equal ($AB \neq BA$).

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

In-class exercise: 1 min.

$$\begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

Matrix Multiplication: AB and BA are NOT Equal

1. AB exists does not imply that BA exists.

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, AB = \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix},$$

BA does not exist.

2. Even if both AB and BA exist, they are generally not equal ($AB \neq BA$).

Example:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Question: Do there exist A, B such that $AB = BA$?

Missing Property of Matrix Multiplication

What property does matrix multiplication NOT have?

[Compared to: scalar product; vector inner product]

Answer: matrix multiplication is NOT commutative.

In Chinese: 矩阵乘积不是可交换的。

Lemma for Property 1

Lemma (Distributive Property of Matrix-Vector Multiplication)

Let $A \in \mathbb{R}^{m \times n}$ be a matrix, and let $\mathbf{b}, \mathbf{c} \in \mathbb{R}^{n \times 1}$ be column vectors. Then matrix-vector multiplication distributes over addition:

$$A(\mathbf{b} + \mathbf{c}) = A\mathbf{b} + A\mathbf{c}.$$

Suppose $A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}$

then $A(\mathbf{b} + \mathbf{c}) =$

row form
of matrix
- vec

$$\begin{bmatrix} \mathbf{a}_1^T(\mathbf{b} + \mathbf{c}) \\ \vdots \\ \mathbf{a}_m^T(\mathbf{b} + \mathbf{c}) \end{bmatrix} =$$

$\mathbf{a}_i^T(\mathbf{b} + \mathbf{c}) = \mathbf{a}_i^T\mathbf{b} + \mathbf{a}_i^T\mathbf{c}$
same linearity of inner product -

$$\begin{bmatrix} \mathbf{a}_1^T\mathbf{b} + \mathbf{a}_1^T\mathbf{c} \\ \vdots \\ \mathbf{a}_m^T\mathbf{b} + \mathbf{a}_m^T\mathbf{c} \end{bmatrix}$$

$A\mathbf{b} + A\mathbf{c} =$

$$\begin{bmatrix} \mathbf{a}_1^T\mathbf{b} \\ \vdots \\ \mathbf{a}_m^T\mathbf{b} \end{bmatrix} + \begin{bmatrix} \mathbf{a}_1^T\mathbf{c} \\ \vdots \\ \mathbf{a}_m^T\mathbf{c} \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^T\mathbf{b} + \mathbf{a}_1^T\mathbf{c} \\ \vdots \\ \mathbf{a}_m^T\mathbf{b} + \mathbf{a}_m^T\mathbf{c} \end{bmatrix}$$

row form
of matrix-vec

sum
of vec

Lemma for Property 1

Lemma (Distributive Property of Matrix-Vector Multiplication)

Let $A \in \mathbb{R}^{m \times n}$ be a matrix, and let $\mathbf{b}, \mathbf{c} \in \mathbb{R}^{n \times 1}$ be column vectors. Then matrix-vector multiplication distributes over addition:

$$A(\mathbf{b} + \mathbf{c}) = A\mathbf{b} + A\mathbf{c}.$$

Suppose $A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix}$ then $A(\mathbf{b} + \mathbf{c}) = \begin{bmatrix} \mathbf{a}_1^T(\mathbf{b} + \mathbf{c}) \\ \mathbf{a}_2^T(\mathbf{b} + \mathbf{c}) \\ \vdots \\ \mathbf{a}_m^T(\mathbf{b} + \mathbf{c}) \end{bmatrix}$

$A\mathbf{b} + A\mathbf{c} = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} \mathbf{b} + \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} \mathbf{c} = \begin{bmatrix} \mathbf{a}_1^T \mathbf{b} \\ \mathbf{a}_2^T \mathbf{b} \\ \vdots \\ \mathbf{a}_m^T \mathbf{b} \end{bmatrix} + \begin{bmatrix} \mathbf{a}_1^T \mathbf{c} \\ \mathbf{a}_2^T \mathbf{c} \\ \vdots \\ \mathbf{a}_m^T \mathbf{c} \end{bmatrix}$

$\mathbf{a}_i^T(\mathbf{b} + \mathbf{c}) = \mathbf{a}_i^T \mathbf{b} + \mathbf{a}_i^T \mathbf{c}$
inner product distributive rule (linearity)

Matrix Multiplication Property 1: Proof (cont'd)

Property Let $A \in \mathbb{R}^{m \times n}$, $B, C \in \mathbb{R}^{n \times l}$, $\alpha \in \mathbb{R}$

$$(1) A(B + C) = AB + AC$$

Proof of (1): (The others are exercises)

Suppose

$$B = [\mathbf{b}_1, \dots, \mathbf{b}_l], C = [\mathbf{c}_1, \dots, \mathbf{c}_l],$$

then

$$B + C = [(\mathbf{b}_1 + \mathbf{c}_1), (\mathbf{b}_2 + \mathbf{c}_2), \dots, (\mathbf{b}_l + \mathbf{c}_l)].$$

Thus

$$\begin{aligned} \text{Lemma} \leftarrow \text{property of } A \cdot (\text{col form}) \downarrow \quad \textcircled{AB} + \textcircled{AC} &= [A\mathbf{b}_1, A\mathbf{b}_2, \dots, A\mathbf{b}_l] + [A\mathbf{c}_1, A\mathbf{c}_2, \dots, A\mathbf{c}_l] \\ &= [A\mathbf{b}_1 + A\mathbf{c}_1, \dots, A\mathbf{b}_l + A\mathbf{c}_l] = [A(\mathbf{b}_1 + \mathbf{c}_1), \dots, A(\mathbf{b}_l + \mathbf{c}_l)] \\ \text{property of } A \cdot (\text{col form}) \swarrow &= A(B + C) \end{aligned}$$

In-class exercise: 1 min.

Rules for Vector and Matrix Multiplication

3 Rules of Real Number Multiplication:
commutative, associative, distributive,.

In Chinese: 交换律, 结合律, 分配律

	commutative	Associative	Distributive
Real number multiplication	Yes	Yes	Yes
Vector inner product	Yes	N/A	Yes
Matrix multiplication	No	Yes	Yes
...			

N/A means: not applicable.

inner product is

$$\langle \overbrace{(\vec{u}, \vec{v})}^{\text{scalar}}, \overbrace{\vec{w}}^{\text{vector}} \rangle$$
$$= \langle \vec{u}, (\vec{v}, \vec{w}) \rangle$$

$$(AB)C = A(BC)$$
$$= ABC$$

Part III Transpose and Matrix Multiplication

15 mins

Matrix Transpose

Definition (Transpose)

Let $A = (a_{ij})_{m \times n}$, then the transpose of A is the matrix $B = (b_{ji})_{n \times m}$, where

$$b_{ji} = a_{ij} (i = 1, \dots, m, j = 1, \dots, n).$$

Notation: $B = A^T$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad A^T = [1, 2, 3]$$

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 5 & -1 & 0 \end{bmatrix}, \quad A^T = \begin{bmatrix} 1 & 5 \\ 0 & -1 \\ 3 & 0 \end{bmatrix}$$

2×3 3×2

Exercise: Transpose By Symbols

Exercise

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} \mathbf{a}^{(1)} \\ \mathbf{a}^{(2)} \\ \vdots \\ \mathbf{a}^{(m)} \end{bmatrix},$$

$\xrightarrow{1 \times n}$
 $\xrightarrow{1 \times n}$

where $\mathbf{a}^{(i)} = [a_{i1} \ a_{i2} \ \dots \ a_{in}]$.

In-class exercise: 2 min.

Exercise: write A^T using a_{ij} and $\mathbf{a}^{(1)}, \dots, \mathbf{a}^{(m)}$.

Correct answer:

$$A^T = \begin{bmatrix} a_{11} & \dots & a_{m1} \\ a_{12} & \dots & a_{m2} \\ \vdots & \ddots & \vdots \\ a_{1n} & \dots & a_{mn} \end{bmatrix} = \left[(\mathbf{a}^{(1)})^T, \dots, (\mathbf{a}^{(m)})^T \right].$$

Incorrect answer:

$$A^T = \underbrace{\left[\underbrace{\mathbf{a}^{(1)}}_{1 \times n}, \dots, \underbrace{\mathbf{a}^{(m)}}_{1 \times n} \right]}_{1 \times (mn)}, \quad \text{Impefect} \quad \left[\mathbf{a}^{(1)T}, \dots, \mathbf{a}^{(m)T} \right]$$

Properties of Transpose

Properties

Let $A, B \in \mathbb{R}^{m \times n}$, $\alpha \in \mathbb{R}$, then

(1) $(A + B)^T = A^T + B^T$. sum

(2) $(\alpha A)^T = \alpha A^T$. scalar product.

(3) $(A^T)^T = A$.

(4) $(AB)^T = B^T A^T$

$$(AB)^T \neq A^T B^T$$

Property 4: Proof

Properties Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times l}$.

(4) $(AB)^T = B^T A^T$

Proof of (4): (The others are exercises)

$$A = (a_{ij})_{m \times n}, B = (b_{ij})_{n \times l}.$$

The (i, j) -entry of AB is $\sum_{k=1}^n a_{ik} b_{kj}$.

The (j, i) -entry of $(AB)^T$ is the same as the (i, j) -th entry of AB , which is $\sum_{k=1}^n a_{ik} b_{kj}$.

(j, i) -entry of $B^T A^T$ is $\sum_{k=1}^n (B^T)_{jk} (A^T)_{ki} = \sum_{k=1}^n b_{kj} a_{ik} = \sum_{k=1}^n a_{ik} b_{kj}$.

Thus, (j, i) -entry of $(AB)^T$ and $B^T A^T$ are the same.

Symmetric Matrix

Definition (Symmetric Matrix)

If a matrix $A \in \mathbb{R}^{m \times n}$ satisfies $A = A^T$ we call A is symmetric. *then $m=n$.*

$\downarrow$ $\leftarrow$
 $m \times n$ $n \times m$

Examples:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix},$$

$$A = \begin{bmatrix} 1 & 3 & -5 \\ 3 & 4 & 7 \\ -5 & 7 & 5 \end{bmatrix}$$

Non-examples:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 3 & 4 & 1 \end{bmatrix}.$$

Matrix Power of Square Matrices

Special case: multiply the same matrix.

What is A^2 ? Easy, should be $\underbrace{A}_{m \times n} \cdot \underbrace{A}_{m \times n}$.

valid
only if $m=n$
(square matrix)

What does A^3 mean?

Two possibilities: $A(A^2)$ or $A^2(A)$.

$$A^p = AAA \cdots A \text{ (} p \text{ factors)}$$

$$(A^p)(A^q) = A^{p+q}$$

$$(A^p)^q = A^{pq}.$$

Self-Product of Non-square Matrices

Judgement: A^2 is NOT a valid expression for a ~~non-square matrix~~ A.

True.

Non-square matrix: m by n matrix where $m \neq n$.

Self-Product of Non-square Matrices

A^2 is NOT a valid expression for a non-square matrix A .

Valid products: (self-products of A)

$$\underbrace{A^T}_{n \times m} \underbrace{A}_{m \times n} \text{ and } \underbrace{A}_{m \times n} \underbrace{A^T}_{n \times m}.$$

$$(A^T A)^2, (A^T A)^3, \dots$$
$$(A A^T)^2, (A A^T)^3, \dots$$

Useful in: optimization, machine learning, etc.

Part IV Valid and Invalid Vector-Vector Products

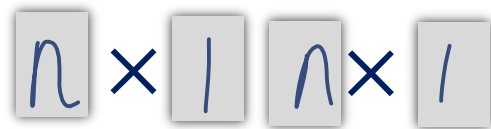
15 mins

Vector-vector products (**same size**)

Want to multiply $u \in \mathbb{R}^{n \times 1}$ and $v \in \mathbb{R}^{n \times 1}$, **two column vectors**.

Which of the following products are valid?
(the notation means **matrix multiplication**)

In-class exercise: 3 min.

$$uv$$


vu

$n \times 1 \quad n \times 1$

$$uv^T$$

$n \times 1 \quad 1 \times n$

✓

$$v^T u$$

~~$1 \times n$~~ $n \times 1$

✓

$$u^T v$$

$1 \times n \quad n \times 1$

✓

$$vu^T$$

$n \times 1 \quad 1 \times n$

✓

Common Confusion

A student's question: Is $u^T v$ an inner product, given $u \in \mathbb{R}^{n \times 1}$ and $v \in \mathbb{R}^{n \times 1}$?

Fact 1:

$u^T v$ equals the inner product of u and v .

Fact 2:

$u^T v$ is the product of u^T and v .

Vector-vector products (**same size**)

	$\mathbf{x}$	$\mathbf{y}$	Product expression	Product type	Reason of valid or invalid
Invalid	$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$\mathbf{x} \mathbf{y}$	Matrix multiplication	"2×1 matrix times 2×1 vector" is not defined
valid	$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$\mathbf{x}^T \mathbf{y}$	Matrix multiplication	<i>Matrix multiplication of 1×2 matrix $\mathbf{x}^T$ and 2×1 matrix $\mathbf{y}$</i>
valid	$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$	$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$	$\mathbf{x} \cdot \mathbf{y}$	dot product	dot product of two 2×1 vector vectors is valid

Vector-vector products (**different size**)

Want to multiply $u \in \mathbb{R}^{m \times 1}$ and $v \in \mathbb{R}^{n \times 1}$, $m \neq n$, two column vectors.

You are allowed to use transpose.

Which products are valid?

In-class exercise: 1-2 min.

$$uv$$

$m \times 1 \quad n \times 1$

~~X~~

$$vu$$

$n \times 1 \quad m \times 1$

~~X~~

$$uv^T$$

$m \times 1 \quad 1 \times n$

$$v^T u$$

$1 \times n \quad m \times 1$

~~X~~

$$u^T v$$

$1 \times m \quad n \times 1$

$$vu^T$$

$n \times 1 \quad 1 \times m$

A diagram illustrating the multiplication of two vectors. On the left, a tall, narrow rectangle represents a column vector of size $m \times 1$. To its right is a short, wide rectangle representing a row vector of size $1 \times n$. An equals sign follows, leading to a large square bracket on the right. Inside the bracket is a smaller rectangle representing the resulting matrix, with a vertical line through its center. To the right of the bracket is a subscript n . Below the bracket is a subscript m .

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1st row $[u_i]$.


$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 \\ u_2 v_1 & u_2 v_2 \\ u_3 v_1 & u_3 v_2 \end{bmatrix}.$$

~~$$\begin{bmatrix} u_1 v_1 + u_2 v_2 \end{bmatrix}$$~~

Vector Outer Product

Definition (Outer Product of Two Vectors)

Let $u \in \mathbb{R}^{m \times 1}$ be a column vector and $v \in \mathbb{R}^{n \times 1}$ be a column vector. The **outer product** of u and v is defined as:

$$uv^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \dots & u_m v_n \end{bmatrix}.$$

Vector Outer Product

Definition (Outer Product of Two Vectors)

Let $u \in \mathbb{R}^{m \times 1}$ be a column vector and $v \in \mathbb{R}^{n \times 1}$ be a column vector. The **outer product** of u and v is defined as:

$$uv^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \dots & u_m v_n \end{bmatrix}.$$

Remark:

The **outer product** of u and v is the **multiplication** of u and v^T .
Don't say "outer product of u and v^T ".

Recall: For two column vectors u and v :

The **inner product** of u and v is the multiplication of u^T and v ,
not "inner product of u^T and v ".

Concluding Section



Summary Today

Write your summary below.

One sentence summary:

Detailed summary:

Summary Today

Instructor's summary

One sentence summary:

We learned (deeper) matrix multiplication, transpose.

Detailed summary:

—Matrix multiplication and properties:

Two rules: Row times column; Dimension match.

Properties: Non-commutative; associative, distributive.

—Transpose.

Transpose and matrix operations (sum, product, partition).

Valid vector-vector products: inner product, outer product.